ICT 2403

VIDEO-BASED ROAD SURFACE DAMAGE DETECTION USING IMAGE PROCESSING

VIDEO ACQUISITION METHOD

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1. Video Capture Device

The road video footages were recorded using an Apple iPhone 13 Pro Max with the rear-facing 12 MP wide-angle camera (26 mm equivalent), featuring an $f/1.5$ aperture, 1.9 µm pixel size, Sensor-shift Optical Image Stabilization (OIS) and Dual Pixel Phase Detection Autofocus (PDAF). The device was chosen due to its available to the project team and the integrated camera is capable of recording stable, well-focused and high-resolution video with good low-light performance, making it suitable for road surface analysis.

2. Recording Parameters and Their Significance

a. Resolution

We choose the video resolution to be 1080p Full HD (1920 x 1080). The phone supports 4K, 1080p, and 720p, but we choose 1080p Full HD (1920 × 1080) for this project because it provides a good balance between video clarity, processing speed, and storage efficiency. The 4K has higher detail, and it produces very large files that are harder to store and process. Then 720p has smaller file sizes but may miss small details like minor cracks or uneven surfaces. But 1080p is sufficient to capture clear and detailed footage without creating unnecessary storage or processing challenges.

b. Frame Rate

The recording speed of the phone can be 24 fps, 30 and 60 fps, but we relied on filming in 30 fps (frames-per-second) due to it being enough frames in order for one capture a smooth action (that you need from the road surface) while keeping the file size/file processing low. The frame rate (fps) is defined as the number of recorded individual frames per second in a video. The higher fps means motion is more seamless, giving a clearer picture and reducing the need to adjust or refocus when capturing scenes of moving objects (like cars down the highway) or animated sequences. 30 fps is a good compromise between smoothness and file size. So that the video is clear, easy to analyze and will not create extremely large files. Which is why we shot our videos at 30 fps.

c. Duration of Video

The videos were recorded with durations ranging from minimum of 18 seconds to a maximum of 35 seconds. Every video covers a specific segment of the road, capturing enough footage to analyse the cracks, potholes, alligator cracking, edge cracking, ravelling and pedestrian crossing marking removal. The duration selected makes sure that the whole road segment that is important has been recorded without creating unnecessary lengthy videos, which allows for effective processing, analysis, and storage.

3. Environmental Conditions

The recordings were performed in the daytime and under good natural light conditions. There was enough natural light at the time of recording (during daytime) for good visibility of the road and to produce better quality videos. The weather was dry so road surfaces weren't wet and that meant surface features such as cracks, potholes, alligator cracking, edge cracking and irregularities were visible. No shadow was present during the recording. Trees existed in some areas that shadowing was recorded from spaces without or limited effect of shade on the trees. Also, streets with low traffic vehicles were chosen for no obstacles and a continuous video recording. These weather conditions contributed to clear and stable videos, which were suitable for accurate road surface analysis.

4. Camera Motion and Placement

The video was recorded using a smartphone. The height of camera is 1.2 meters from the road surface. The phone was mounted at an approximate 45 degree angle, allowing the frame to capture both the road surface and a portion of the area ahead. This angled orientation helped in maintaining proper context while still focusing on road surface conditions. The recording was done while walking with the phone held steadily by hand, rather than being mounted on a vehicle. This approach facilitated greater flexibility in movement, while guaranteeing full coverage of targeted road sections.

5. Site Selection

The video recordings were mainly conducted in rural areas. Rural roads were selected because they generally have low traffic conditions and more open space, which makes video recording easier and safer without Continuous interruptions from vehicles. Due to these advantages, village-side roads were chosen for data collection. Road segments located on the village area was selected and several videos were also recorded near the researcher's village (Team mate's hometown). These roads showed visible surface conditions such as cracks, potholes, alligator cracking, edge cracking, ravelling and pedestrian crossing marking removal, which were suitable for analysis. Some locations were excluded from the final dataset because the weather conditions were not favourable during recording, which could affect video quality. Selecting these specific road segments ensured clear, stable, and reliable footage for the project.

6. Recording Process and Sample Videos

The recording procedure was carried out in a systematic manner to ensure consistent and clear video data. Before recording, the camera settings were checked and set to 1080p resolution at 30 fps. The smartphone was held by hand at a height of approximately 1.2 meters and angled at around 45 degrees toward the road surface. Recording was done while walking slowly, maintaining a steady motion to minimize vibrations. Care was taken to avoid shadows, heavy traffic, and poor lighting conditions during recording. Only road segments with clear visibility and suitable environmental conditions were selected. In total, approximately 15 to 23 videos were recorded. Among them 2 to 3 videos were rejected due to poor clarity and lack of stability. The remaining videos were selected for further analysis to maintain overall data quality.

7. Video Analysis and Dataset Preparation

The process of extracting the frames is converting the recorded video into individual images. Videos consist of series of images every second and retrieving these individual images lead to a detailed review of each image. In the following sections, video frames were analysed to observe road conditions such as cracks and localised elevation differences on the road surface. Frame extraction simplifies video data in to images data. The frame extraction extracts the captured video into image frames. As a video is usually composed of frames (captured in a rate of per frame), we simply decompose them for fine-grained analysis. In this research, images were captured from the videos to analyse the conditions inside road surfaces including crack and bump. One of the main goals of frame extraction is to convert video data into image data, which can simplify processing and analysis.

8. Data Annotation Process

The data annotation process was carried out manually, as accurate labeling is essential for proper evaluation and analysis. After frame extraction, the selected images were carefully reviewed, and road surface damages were manually annotated. A basic image editing or labeling tool was used to mark and label the visible damage areas on each frame. The damages were categorized into predefined classes such as cracks, potholes, and uneven road surfaces. Manual annotation ensured that each damage type was clearly and accurately labeled, improving the reliability of the dataset for further analysis and evaluation.

9. Ethical and Safety Considerations

Ethical and safety considerations were carefully followed during the data collection process. The videos were recorded in such a way that no identifiable individuals were captured, ensuring privacy and ethical compliance. Recording was carried out on low-traffic roads to avoid disrupting vehicles and pedestrians. The researcher maintained safe recording practices, paying attention to surroundings while recording and avoiding any actions that could pose a risk to road users or personal safety. These measures ensured that the data collection process was conducted responsibly and ethically.